# MTHS120: Mock Examination

Week 2: Formal Convergence and Limit Laws

**Time Allowed:** 90 Minutes | **Total Marks:** 60

**Instructions:**

- Answer all questions in the spaces provided on this paper.
- Show all necessary algebraic working and scratchpad inequality derivations.
- Fully justify your choice of cutoff milestone $N$ in formal proofs.

## Part A: Conceptual Foundations & Quantifiers (Easy)

*This section tests your understanding of the formal $\epsilon$-$N$ definition and quantifier order.*

1. **[5 marks]** Explain why swapping the order of the first two quantifiers in the formal definition of convergence—changing $\forall \epsilon > 0 \ \exists N \in \mathbb{N}$ to $\exists N \in \mathbb{N} \ \forall \epsilon > 0$—fundamentally breaks the definition and only describes constant sequences.

2. **[5 marks]** State the formal logical negation of the statement $\lim_{n \to \infty} a_n = L$ using $\epsilon$-$N$ notation. Explain how this negation is used to prove that a sequence diverges by oscillation.

## Part B: First-Principles $\epsilon$-$N$ Proofs (Medium)

*This section tests your ability to construct rigorous $\epsilon$-$N$ proofs using the scratchpad method.*

3. **[10 marks]** Use the formal $\epsilon$-$N$ definition of convergence to prove that:

$$\lim_{n\to\infty} \frac{3n-1}{n} = 3$$

Show your scratchpad derivation and explicitly state your choice of $N$ in terms of $\epsilon$.

4. **[10 marks]** Prove from first principles ($\epsilon$-$N$ definition) that:

$$\lim_{n\to\infty} \frac{2n^2+n}{n^2+1} = 2$$

Hint: Find a simplified upper bound for the error term $|a_n - 2|$ before choosing your milestone $N$.

## Part C: Structural Theorems & Limit Laws (Hard)

*This section tests your application of Proposition 5, tail invariance, and algebraic limit laws.*

5. **[10 marks]** State Proposition 5 (Property 2: Convergence Implies Boundedness). Provide a clear outline of the "divide-and-conquer" proof strategy used to bound both the finite prefix and the infinite tail.

6. **[10 marks]** Let $(a_n)$ and $(b_n)$ be sequences such that $\lim_{n\to\infty} a_n = K$ and $\lim_{n\to\infty} b_n = L$. Use the Algebraic Limit Laws and the "$\epsilon/2$ trick" to prove that:

$$\lim_{n\to\infty} (a_n - b_n) = K - L$$

Explicitly show how the master cutoff $N = \max(N_1, N_2)$ synchronizes the two individual error budgets.